CHEM 427/527 · SESSION 12 · TUESDAY, NOVEMBER 10, 2026

Peak detection and feature extraction: does this lavender oil meet the pharmacopoeia?

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Drafted with assistance by Claude Code. Reviewed and revised by Matt.

LEARNING OBJECTIVES

By the end of today you can

1. Choose peak-detection settings from a chromatogram’s noise, its narrowest peak, and the smallest peak that must count, and show what a threshold set too low or too high does.
2. Integrate each peak between the lowest points that separate it from its neighbors, and report each component as a percent of the summed area of the peaks counted.
3. Decide whether an oil meets a specification from three replicate runs, their spread, and the distance to the nearest limit.
4. Integrate the same peaks by a second rule, show how far it moves each result, and say whether it changes your call.
5. Write a skill for a step you repeat, check it against a table you built by hand, and decide whether to change it before it reads new data.

FIRST, EVERYONE PULLS

Get session 12 with git pull

```
git pull
```

S12-peak-detection appears, with

- **data/L13A.cdf**: Dataset A, the worked example
- three runs of each oil, such as **data/L19A.cdf**, **L19B.cdf** and **L19C.cdf**
- **data/samples.csv**: each student number's oil
- **data/README.md**: where the runs come from and how to read them

Session 11 is due today

- Upload it by 1:30 pm.
- Fill in your prompt record, then Restart and Run All before you upload.

Nothing is downloaded today: the 91 runs and samples.csv are in the repository.

THE DECISION

Does this lavender oil meet the pharmacopoeia?

Who decides

A quality-control analyst at a supplier or a pharmacy reads a GC-FID run against the European Pharmacopoeia's profile, by area percent: camphor at most 1.2%, linalool 20.0 to 45.0%, and linalyl acetate 25.0 to 47.0%.

Passed wrongly

An oil cut with lavandin or with synthetic linalool is sold as pharmacopoeial.

Failed wrongly

A genuine lot is rejected, and its supplier loses the sale.

Real data: 55 commercial oils, three GC-FID runs each

Dataset A, L13A.cdf, is a 73 min run on a wax column, one point every 0.1 s. The solvent peaks at 4.39 min and 1,413 mV. Linalool gives the largest of the oil's peaks, 124 mV on a baseline near 1 mV. The files name no peak; one oil's GC-MS run confirmed camphor's place. You have one oil's three runs.

FROM WHERE THE BASELINE SITS TO WHICH PEAKS COUNT

What does integrating every peak down to 0 mV do to linalool’s area percent?

Session 11

An impurity’s area percent depended on where the baseline was drawn.

Today

The baseline sits near 1 mV, under dozens of peaks a few mV tall.

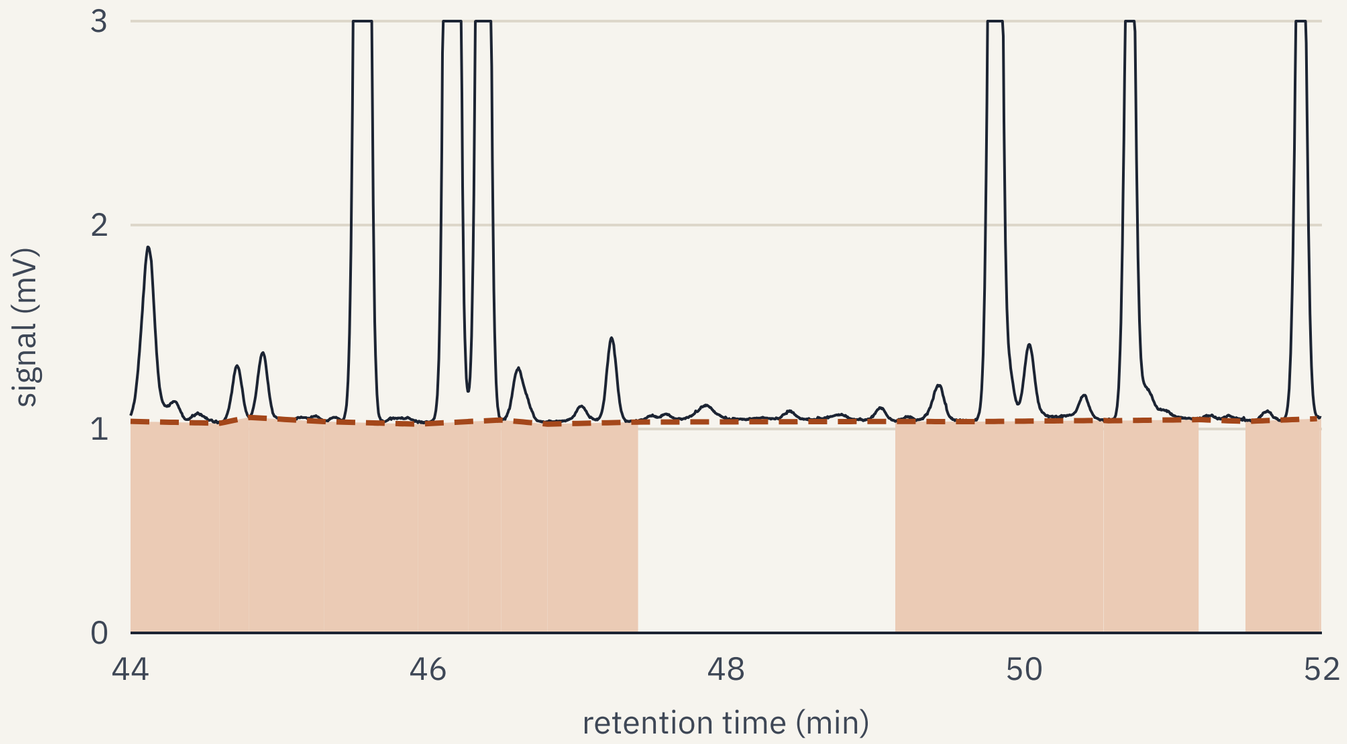
The question

Each peak is integrated down to 0 mV instead of down to that baseline. Where does linalool’s share go?

Does linalool’s area percent go up, go down, or stay the same?

THE ANSWER, ON DATASET A

Each peak gains the slab under it



The drop-line baseline, near 1 mV

The slab that 0 mV adds

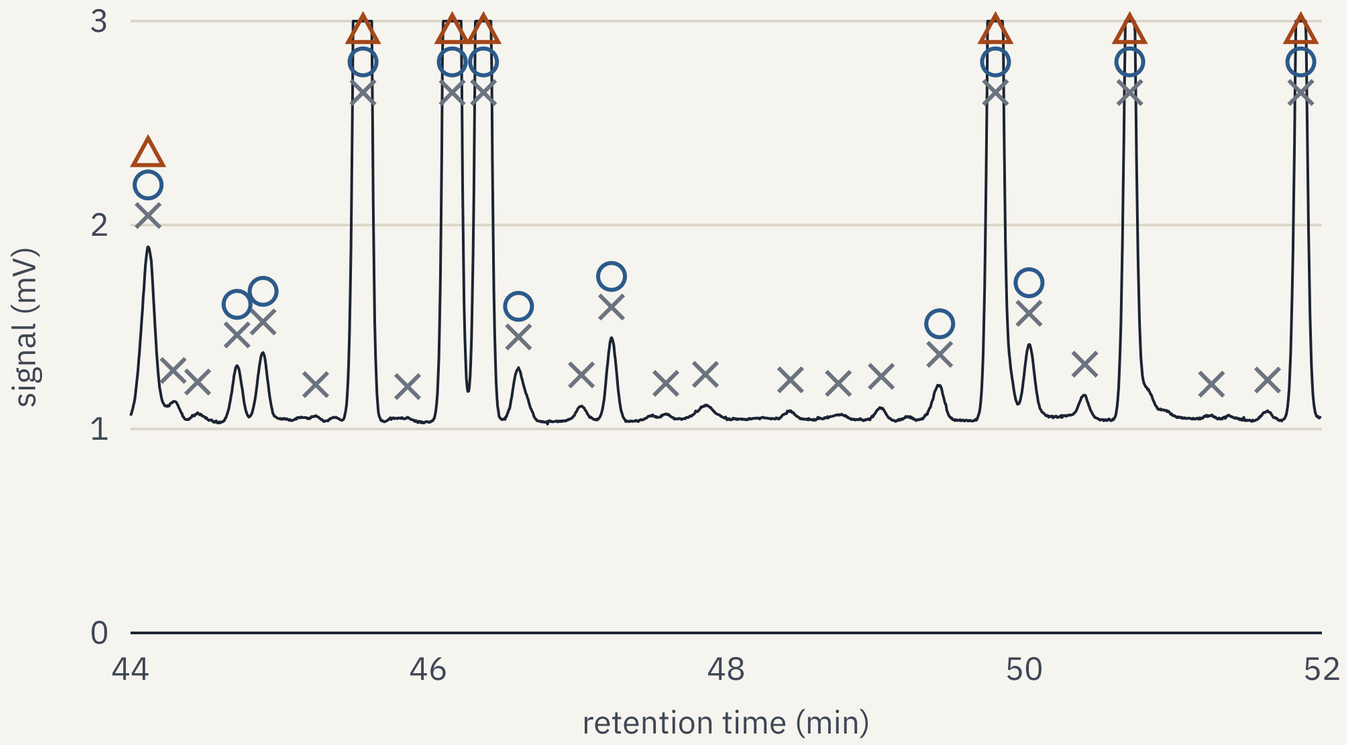
Integrated to	baseline	0 mV
camphor	0.25	0.80
linalool	34.37	17.79
linalyl acetate	35.06	18.35

The table gives each component’s area percent. The total grows faster than linalool’s area, so linalool falls below its 20.0% limit and camphor rises.

A skill that does not say where the baseline is will choose one.

LIVE: THREE PROMINENCES ON DATASET A

Which peaks count toward the total?



- × **5 times the noise, 0.024 mV**: 122 found, 54 kept
- ○ **The disregard-limit setting, 0.124 mV**: 58 found, 53 kept
- △ **Read off the plot, 0.5 mV**: 32 found, missing 21 that the disregard-limit setting keeps

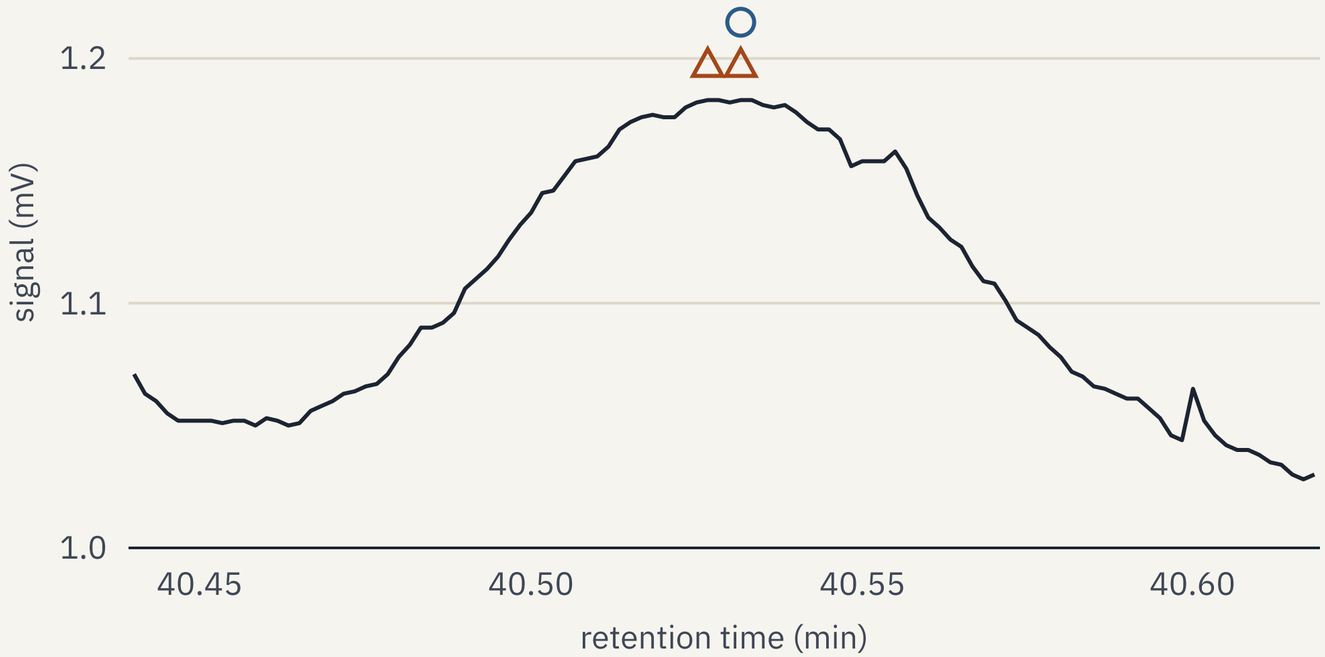
A peak is kept at or above 0.05% of the total area, the disregard limit. From 44 to 52 min the chart marks 26 crosses, 13 circles and 7 triangles.

The data system’s own table, inside the file, lists 38 peaks.

Too low counts noise. Too high misses real peaks.

LIVE: THE SAME DETECTION WITH NO DISTANCE

What keeps one apex per peak?



△ **No distance**: 60 peaks; the peak shown is found twice, at 40.527 and 40.532 min, 0.3 s apart

○ **Distance 35 points**: one apex

The distance is the fewest points allowed between two apexes.

Section 1 sets every setting from two numbers in the run: its noise and its narrowest peak. Your oil's numbers differ.

From what number in the data does the notebook set the distance?

ASSIGNMENTS

One oil each, by student number

#	Oil	#	Oil	#	Oil	#	Oil
1	L6	8	L51	15	L34	21	L19
2	L31	9	L50	16	L25	22	L36
3	L37	10	L3	17	L38	23	L12
4	L44	11	L18	18	L16	24	L29
5	L55	12	L11	19	L43	25	L23
6	L47	13	L1	20	L14	26	L21
7	L39	14	L9				

Your oil is beside your student number from session 4 in data/samples.csv. In Parameters, set **SAMPLE**, such as SAMPLE = "L19", and run the Parameters cell. Section 1 is the worked example on Dataset A. Section 2 is your oil, by hand and by the skill, and section 3 compares the two integration rules.

AI PRACTICE 7 OF 8, NEW TODAY: A SKILL.MD FOR A REPEATED STEP

A file of instructions, run as a command

Live, on Dataset A, in VS Code opened on the repository’s folder

1. In the Explorer, make .claude/skills/peak-report/SKILL.md and paste the template from section 2.

2. Fill the four blanks with section 1’s values: 0.124 mV, 17.6 points, 35 points and drop. Save.

3. In the Claude Code prompt box, type / and choose Clear conversation. Type /skills and find peak-report in the list.

4. Type the command, with Dataset A’s path:

```
/peak-report sessions/S12-peak-detection/data/L13A.cdf
```

What the template states as numbers

- The stretch searched, 5.8 to 69.5 min.
- Every window, in points: 300 for a bound’s search, 180 for the baseline test.
- How each bound is searched.
- Where the baseline is.

A wording that the assistant can read two ways is a step to change.

THE CHECK, ON DATA WHOSE ANSWER YOU KNOW

Does the skill reproduce your hand-built table?

Dataset A, area percent, drop-line at 0.124 mV

Component	By hand	The skill	Data system
camphor	0.25	?	0.25
linalool	34.37	?	35.16
linalyl acetate	35.06	?	35.86

The data system’s column comes from its own list of 38 peaks. It is a third reading, not the answer.

How to read the report

- Its first line: the parameters and the rule it used.
- Its three values against yours, within 0.01 points.
- Anything it took from outside SKILL.md.

A match counts only if SKILL.md’s own text made it. A larger difference points at a step to change before the skill reads your runs.

SECTION 2: THE SKILL ON YOUR RUNS, THEN TASK 3

Your settings, your table, your record

Known answer first, new data second

1. Check the skill on Dataset A. Your prompt is the record’s first line.
2. TASK 2: your settings, from run A’s printout.
3. Put your three numbers in SKILL.md, save, and run /peak-report on runs A, B and C.
4. TASK 3: your own table, in a code cell you add.
5. Fill the record’s other two lines.

The record, a markdown cell

```
**Prompt record**
- **My prompt:**
- **The skill's Dataset A values against my hand-built ones, and my runs' against TASK 3's:**
- **What I changed in SKILL.md, or what I verified (the report's parameter line, and anything it took from outside SKILL.md):**
```

TASK 3: a code cell you write that sets **hand**, a table with one row per run, A, B and C, and a column each for camphor, linalool and linalyl acetate.

The three lines of your prompt record count toward execution.

THE JUDGMENT CELL, SECTION 4

Your oil’s call, and one entry

(a) Your answer

- Does your oil meet the profile for camphor, linalool and linalyl acetate?
- Each component’s mean and spread, the nearest limit and your margin, the other rule’s shift, and the data system’s value
- What a wrong call costs either way

(b) One judgment entry

- Your prompt record is the source.
- What you did: changed, rejected, or verified
- The evidence: the skill’s Dataset A values against your hand-built ones

Which of the three distances, the margin, the spread or the rule’s shift, would make you ask for a fourth run of your oil?

SECTION 1 IS READ AND RUN. THE SKILL AND SECTION 2 ARE YOURS.

Go, then submit

Submit

Fill in your prompt record, then Restart, Run All, save, and upload activity.ipynb by **1:30 pm Thursday, November 12**.

Scoring, 0 to 3

1 if it runs after a restart and holds your prompt record, **0** if not.

0 to 2 for the judgment cell.

Help

Ask Matt, Lucas, or Chris in the room.

The third hour is an office hour.